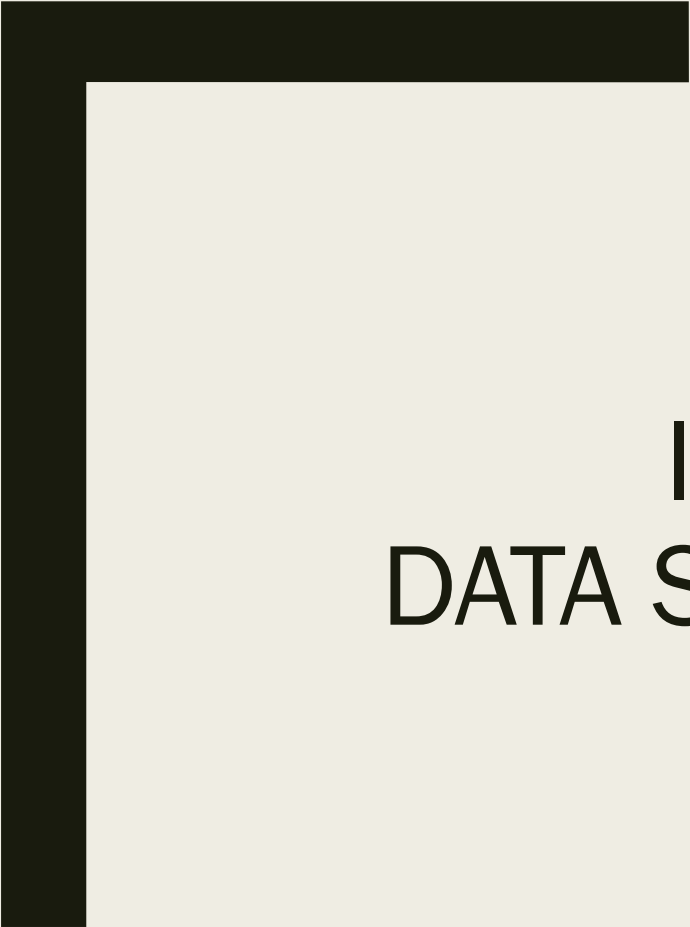
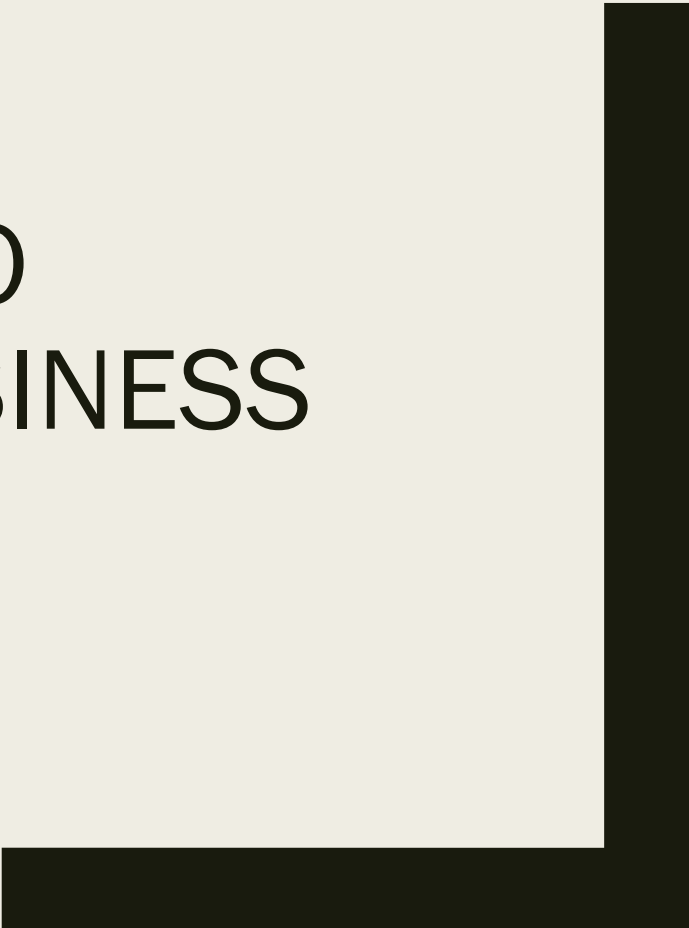




INTRODUCTION TO DATA SCIENCE FOR BUSINESS

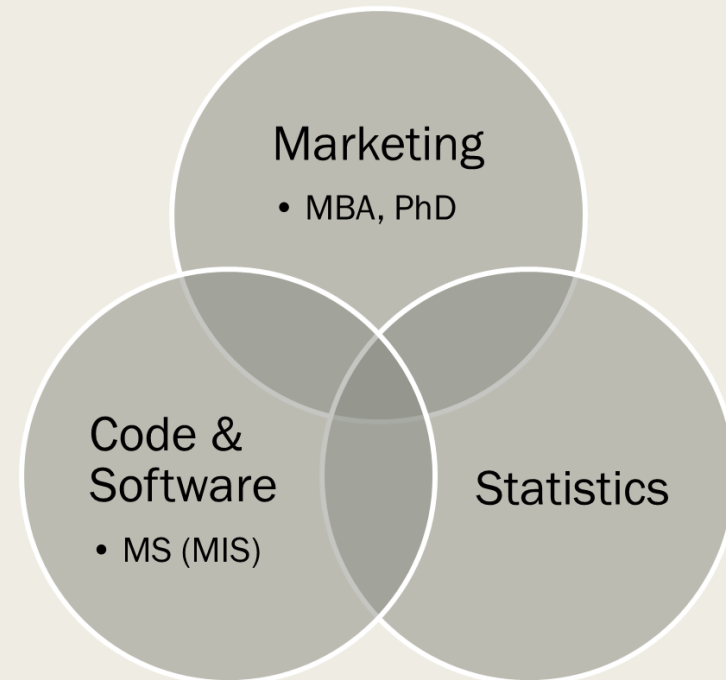


PERSONAL INTRODUCTION

Vishal Lala



- Tenured Professor, Pace University
- Data Science Educator and Consultant



Teaching

- Predictive Analytics
- Data Science
- Data Visualization
- Multivariate Analysis
- Marketing Research
- Digital Marketing

Research

- Consumer Decision Making
- Intergenerational influence
- Consumer-Technology Interaction
- Consumer Generated Content

View list of publications on [Google Scholar](#)

And, when I am not at working..

- Hanging with Rohan and Nikhil
- Working out
- Swimming
- Running
- Playing volleyball
- Playing chess

DATA SCIENCE FOR BUSINESS

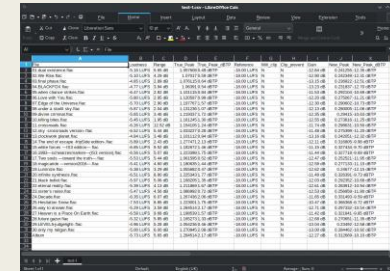
What is Data Science?

- The process of extracting meaningful insights and knowledge from structured and unstructured data using scientific methods, algorithms, and systems.

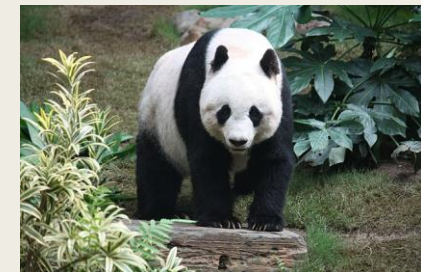
Data

Data is all around us. They differ in their readiness for use

- Raw numbers.
 - *E.g., store transactional data*
- Can be reduced to data
 - *Text, Image, Audio, Video.*
- Characteristics of objects
 - *Car, smartphone, website, thermostat*
 - *Weather, sun, moon*



A screenshot of a data table, likely from a spreadsheet application. The table has multiple columns with headers such as 'Date', 'Time', 'Location', 'Status', 'Value', 'Unit', 'Category', and 'Notes'. The rows contain numerical data, including dates, times, and various values, some of which are highlighted in blue.

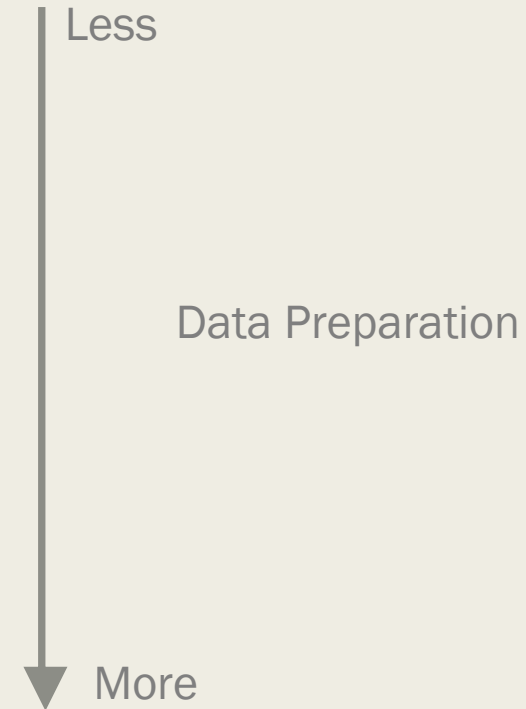


Data Differ in Ease of Use

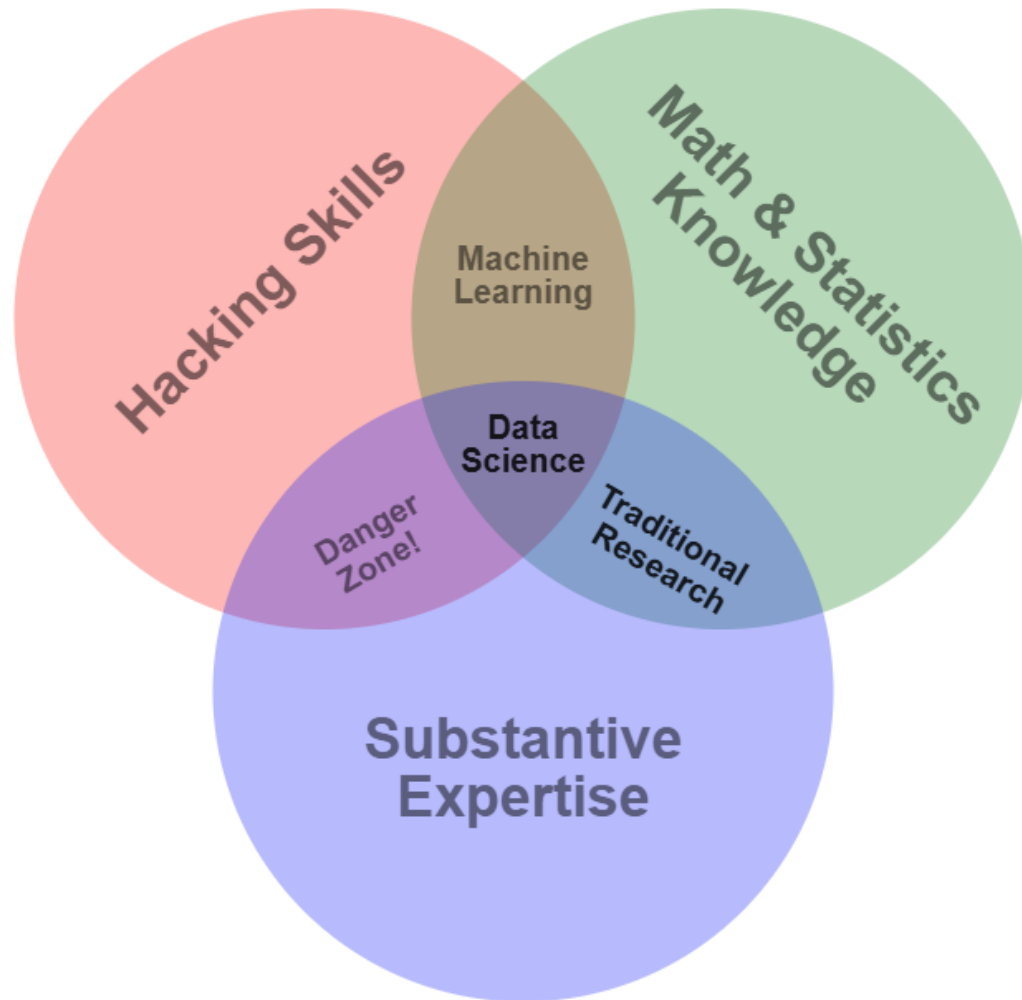
- Some are ready to use. E.g., spreadsheet
- Need to be converted to a usable form. E.g., image
- Need to be gathered or collected. Access API, get permission, gather with a survey
- Messy. Data that is missing, incomplete, outdated, or unreliable.

Data Differ in Ease of Use

- [Tidy](#) spreadsheet.
- [Messy](#) spreadsheet
- PDF
- Text ([Words](#) to [Numbers](#))
- Picture ([Grayscale photo](#) to [Numbers](#))
- Audio
- Video

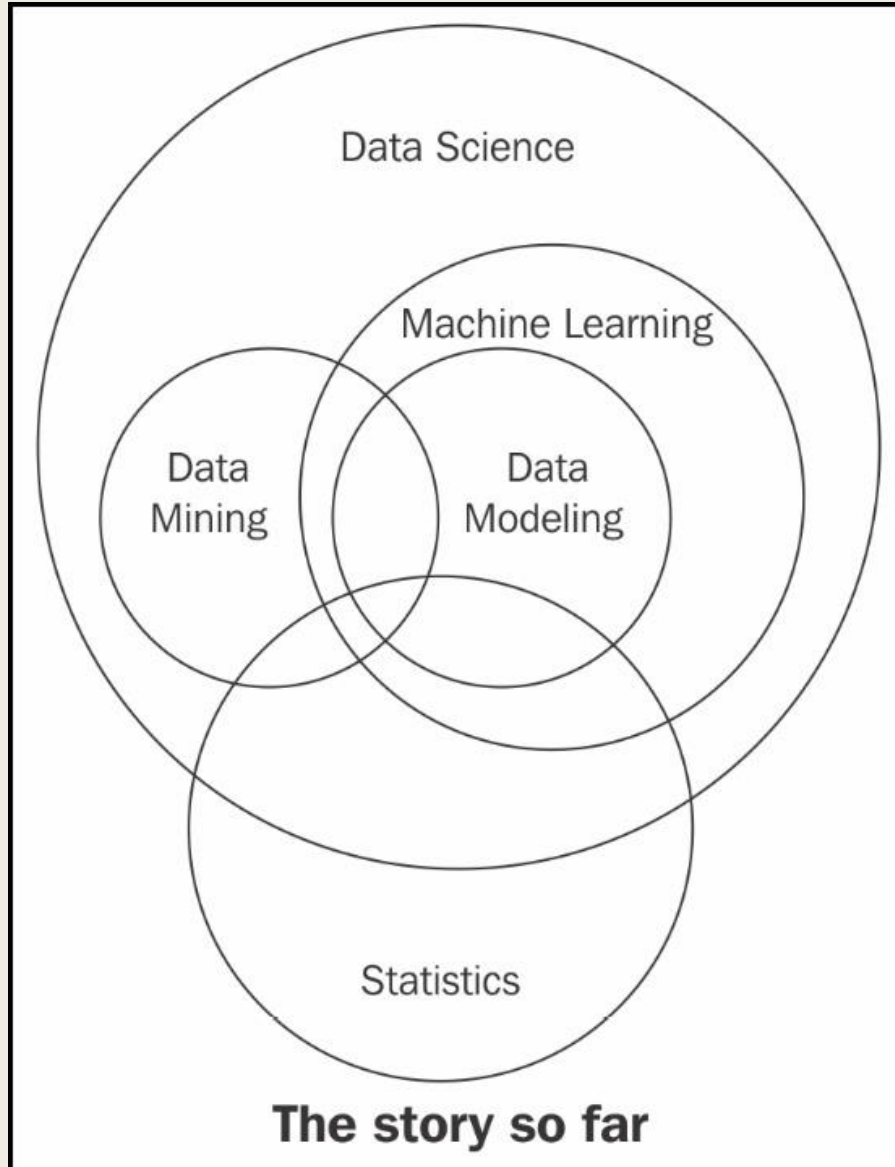


The Data Science Venn Diagram



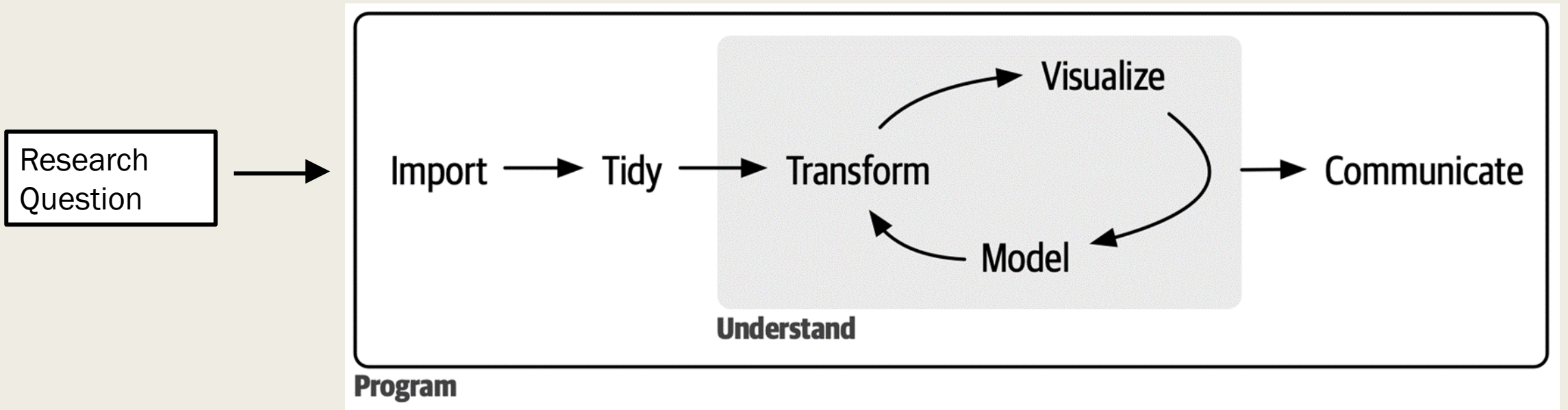
Skills of a Data Scientist

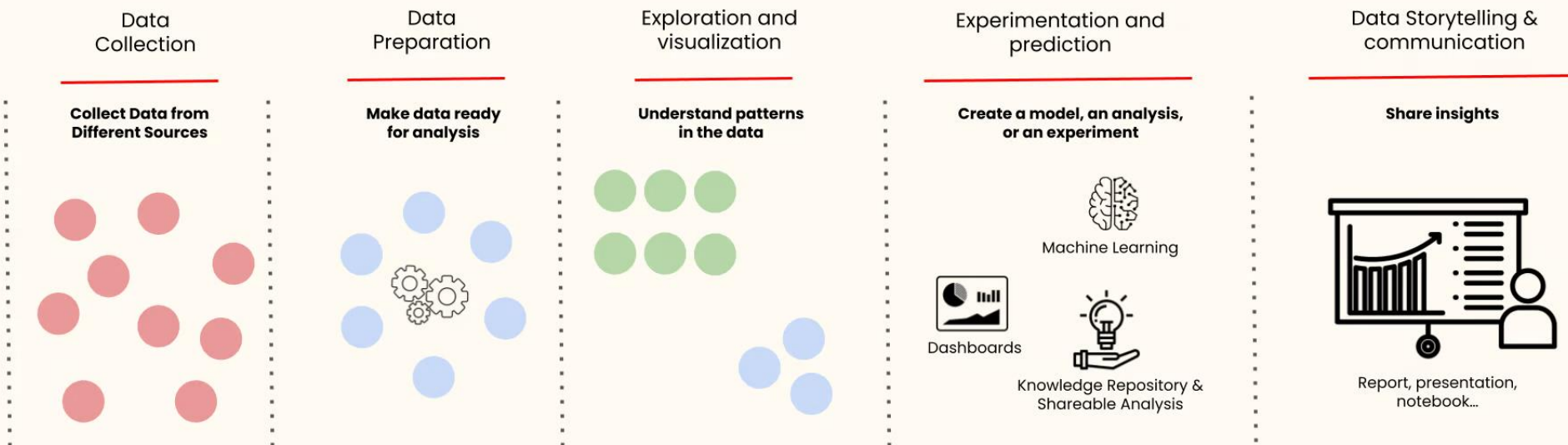
Source: [Drew Conway](#)



Where does Data Science Fit

[Source](#)





[Source](#)

What is Machine Learning?

A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

Tom Mitchell, 1997

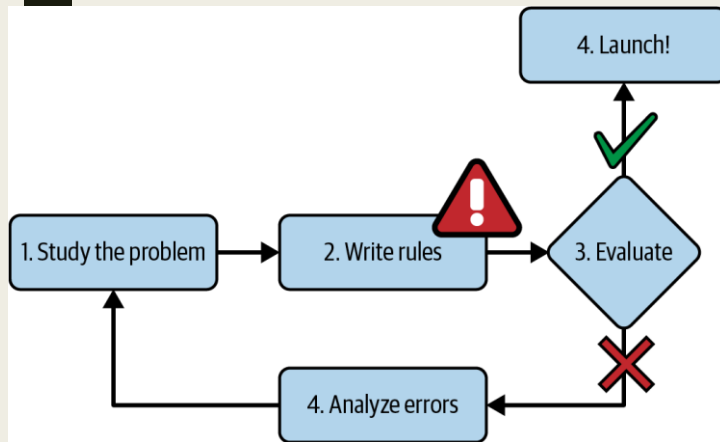
Example:

- Task T : Email classification
- Experience E : Past labeled emails (spam or not)
- Performance P : Accuracy on new emails

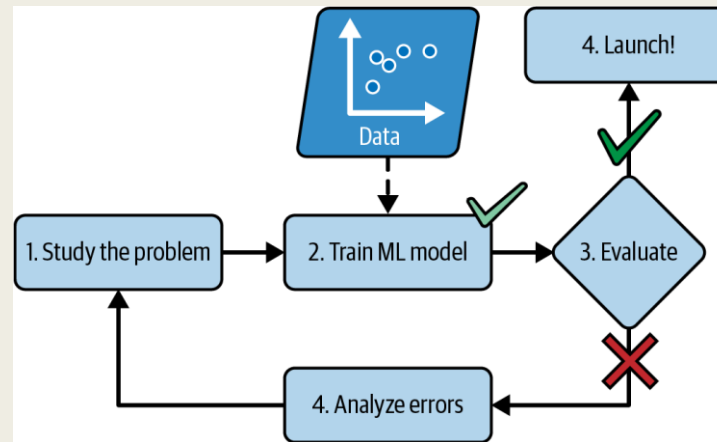
Simple Rules-Based Systems

- Early AI used rules-based systems to mimic human intelligence
- Consider a set of rules for approving loan applications
 - *If FICO > 680 and Credit History > 3 years, then Approve*
 - *If FICO > 680 and Credit History < 3 years, then Request more information*
 - *If FICO < 680, then Reject*
- So, why use Machine Learning?

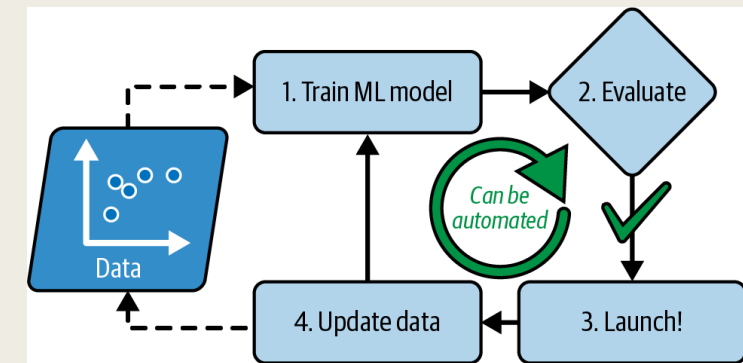
Why use Machine Learning?



Rules-based System



Machine Learning



Machine Learning Adapting to Change

Source: Geron (2022)

Applications

- Medical image diagnosis (e.g., breast cancer detection)
- Credit scoring
- Predict loan default
- Predict likelihood of clicking an ad
- Customer churn prediction
- Predictive policing
- Resume screeners
- Customer segmentation
- Fraud detection
- Image compression
- Identify social influencers
- Spot network vulnerabilities
- Sentiment analysis
- Discover topics in documents
- Discover associations in transaction data

Supervised Machine Learning

- Medical image diagnosis (e.g., breast cancer detection)
- Credit scoring
- Predict loan default
- Predict likelihood of clicking an ad
- Customer churn prediction
- Predictive policing
- Resume screeners

Unsupervised Machine Learning

- Customer segmentation
- Fraud detection
- Image compression
- Identify social influencers
- Spot network vulnerabilities
- Sentiment analysis
- Discover topics in documents
- Discover associations in transaction data

Predictive Analytics

- Medical image diagnosis (e.g., breast cancer detection)
- Credit scoring
- Predict loan default
- Predict likelihood of clicking an ad
- Customer churn prediction
- Predictive policing
- Resume screeners

Data Science for Business

- Customer segmentation
- Fraud detection
- Image compression
- Identify social influencers
- Spot network vulnerabilities
- Sentiment analysis
- Discover topics in documents
- Discover associations in transaction data

&

- Forecasting Weather
- Predicting Stock Price
- Image Classification

Predictive Analytics

- Medical image diagnosis (e.g., breast cancer detection)
- Credit scoring
- Predict loan default
- Predict likelihood of clicking an ad
- Customer churn prediction
- Predictive policing
- Resume screeners

Data Science for Business

- Customer segmentation
- Fraud detection
- Image compression
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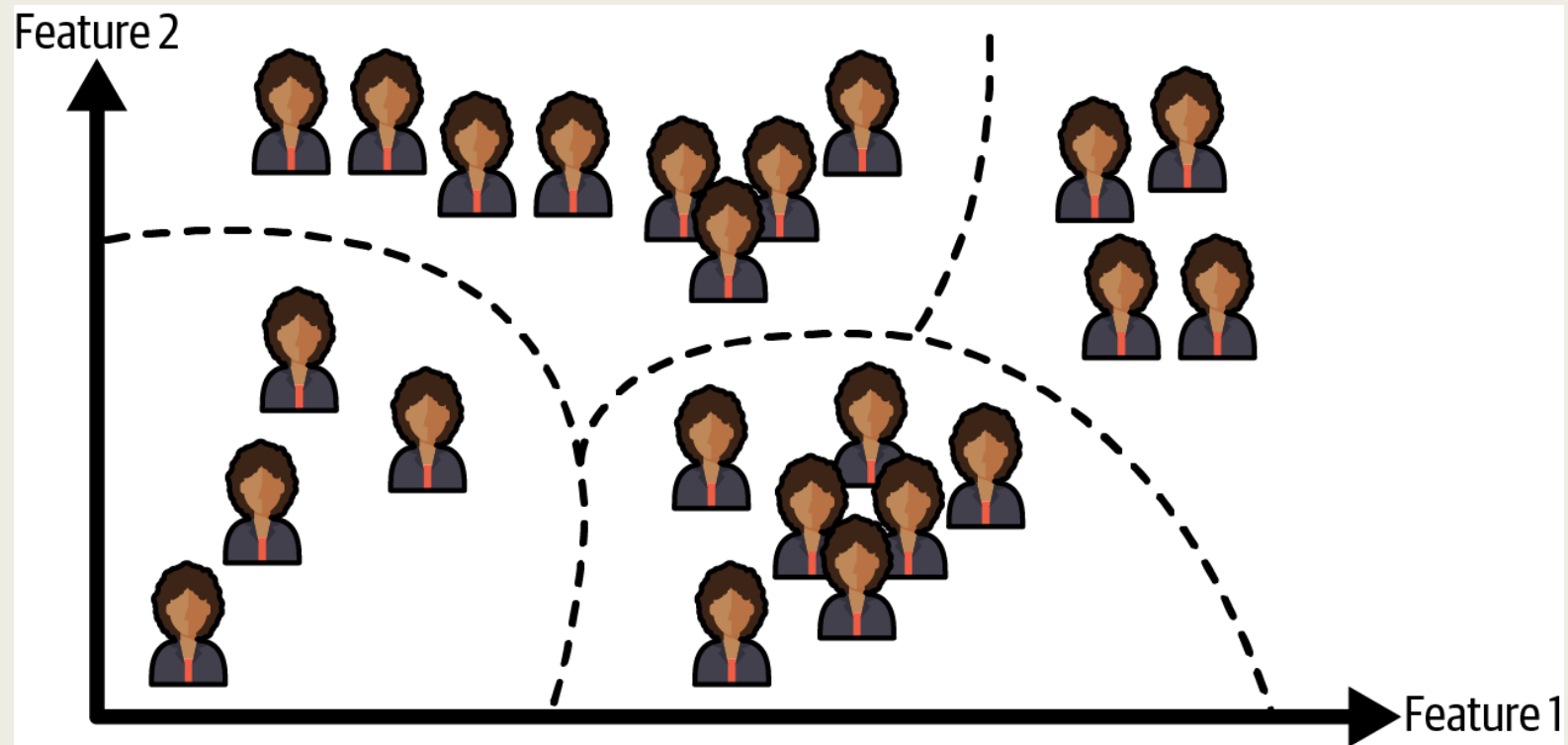
&

- Forecasting Weather
- Predicting Stock Price
- Image Classification

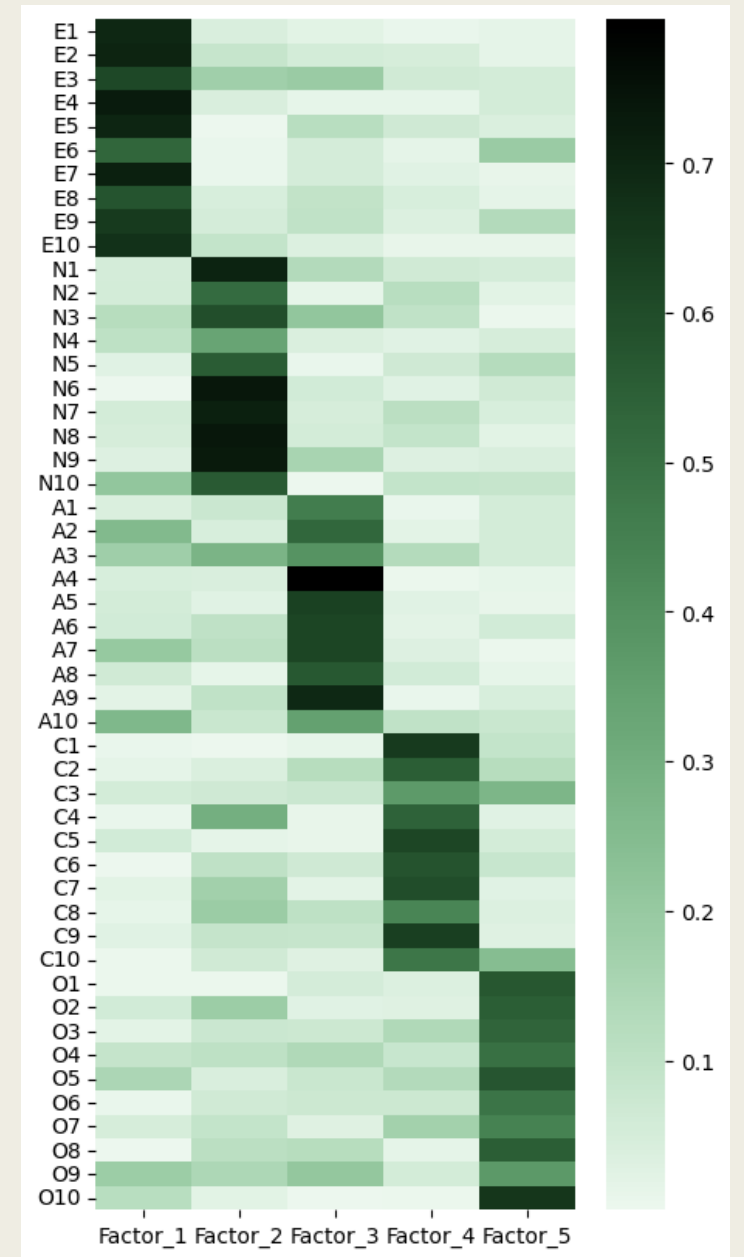
Key Topics

- Clustering
- Dimension Reduction
- Natural Language Processing
- Time Series
- Association Rules
- Recommendation System
- Network Analysis
- Deep Learning

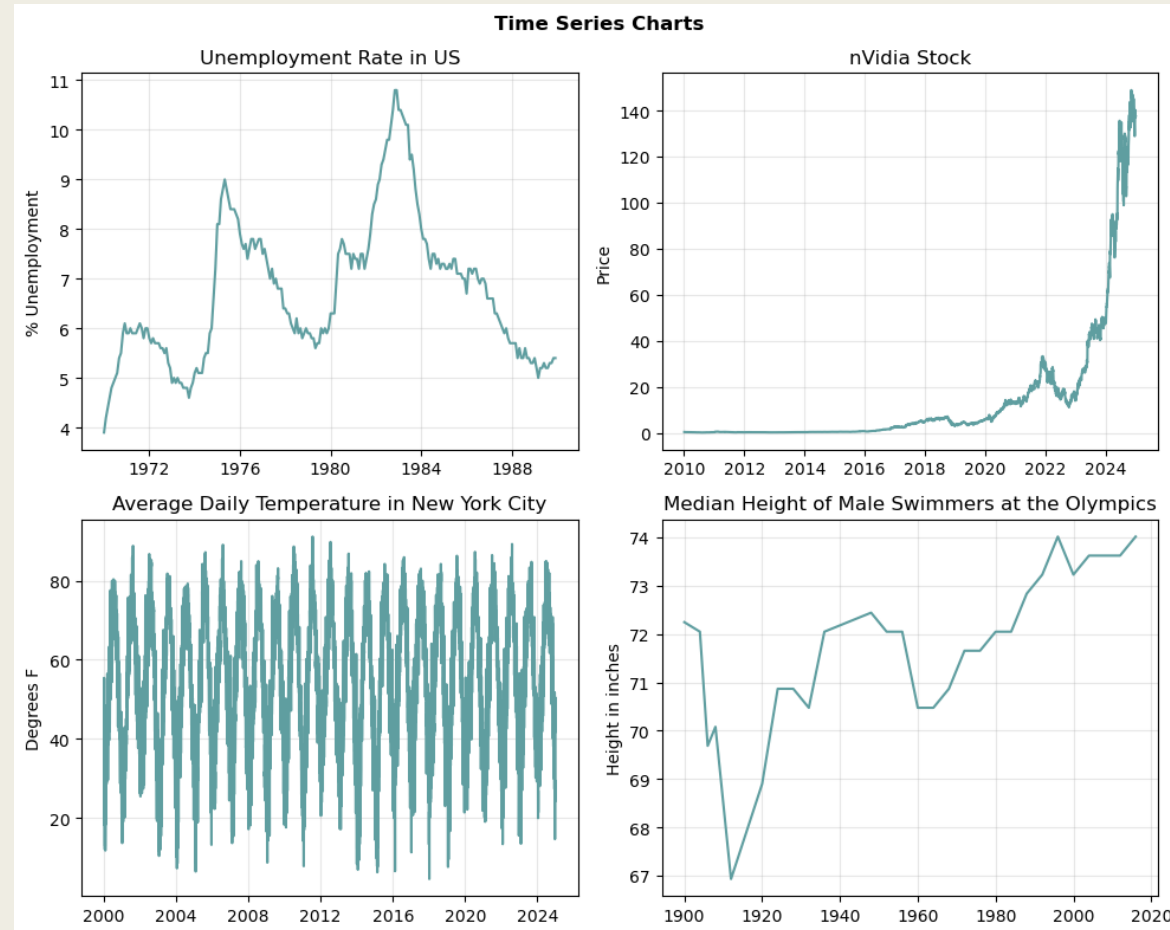
Clustering



Dimension Reduction



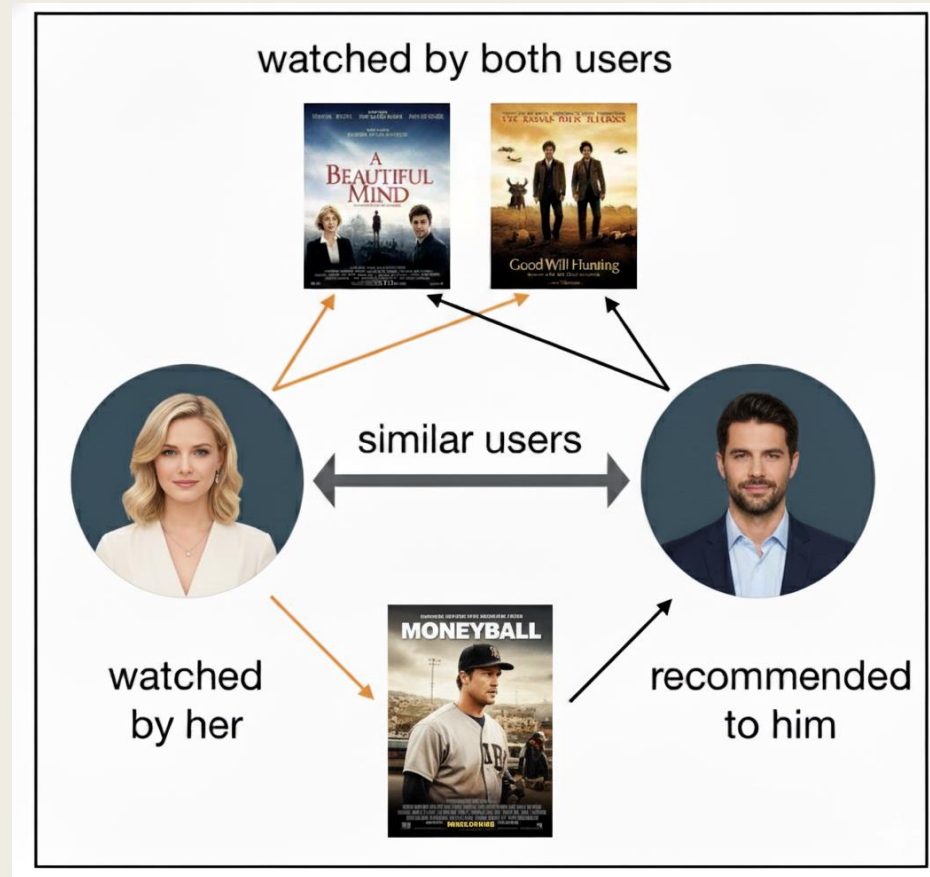
Time Series



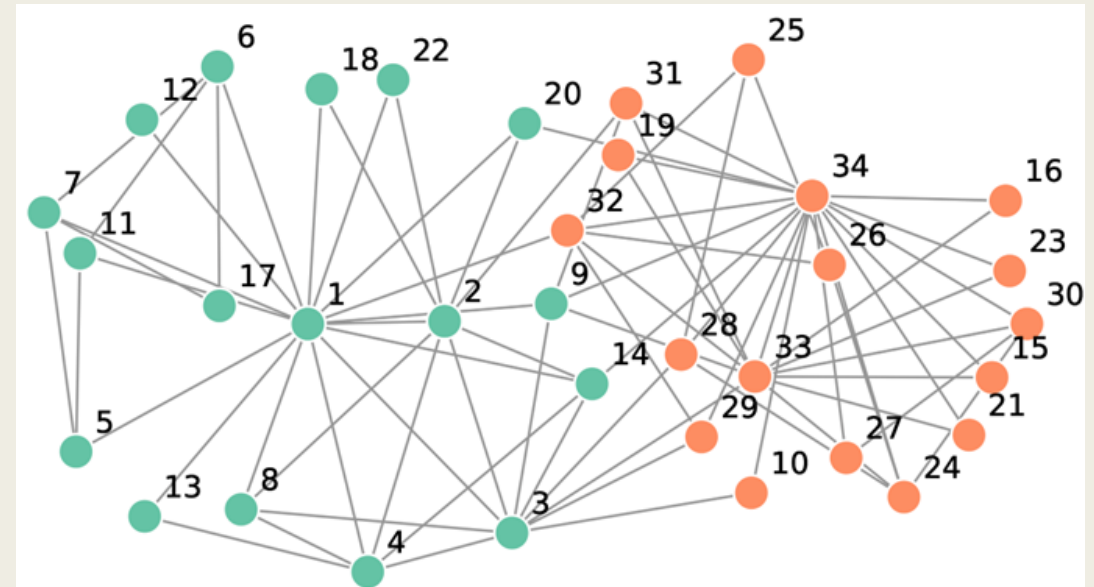
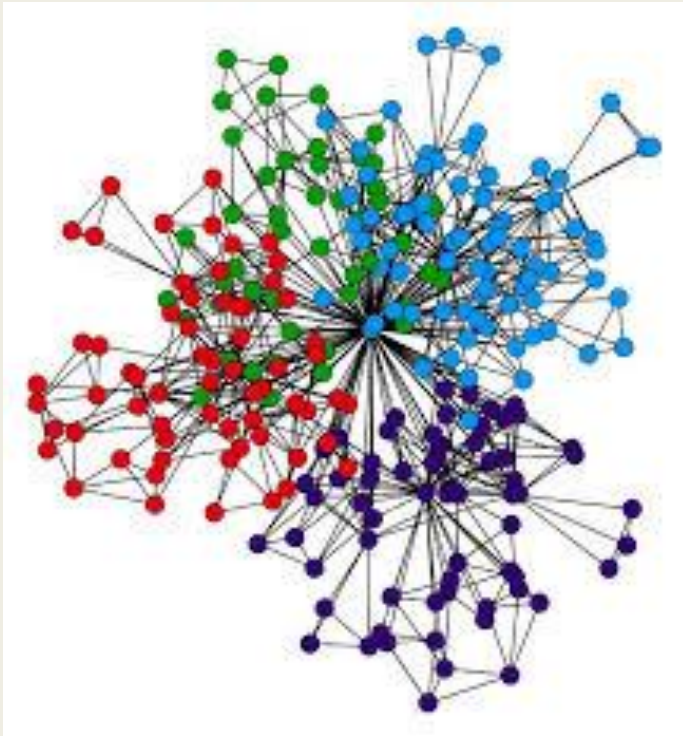
Association Rules

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift
920	(Fossil Watch)	(iPhone 15 Pro Max)	0.03352	0.03400	0.00316	0.094272	2.772708
921	(iPhone 15 Pro Max)	(Fossil Watch)	0.03400	0.03352	0.00316	0.092941	2.772708
1127	(Instant Pot Duo 7-in-1)	(KitchenAid Stand Mixer)	0.03376	0.03736	0.00332	0.098341	2.632260
1126	(KitchenAid Stand Mixer)	(Instant Pot Duo 7-in-1)	0.03736	0.03376	0.00332	0.088865	2.632260
1150	(Instant Pot Duo 7-in-1)	(Vitamix 5200 Blender)	0.03376	0.03508	0.00304	0.090047	2.566915
1151	(Vitamix 5200 Blender)	(Instant Pot Duo 7-in-1)	0.03508	0.03376	0.00304	0.086659	2.566915
943	(Garmin Fenix 7 Watch)	(iPhone 15 Pro Max)	0.03352	0.03400	0.00272	0.081146	2.386635

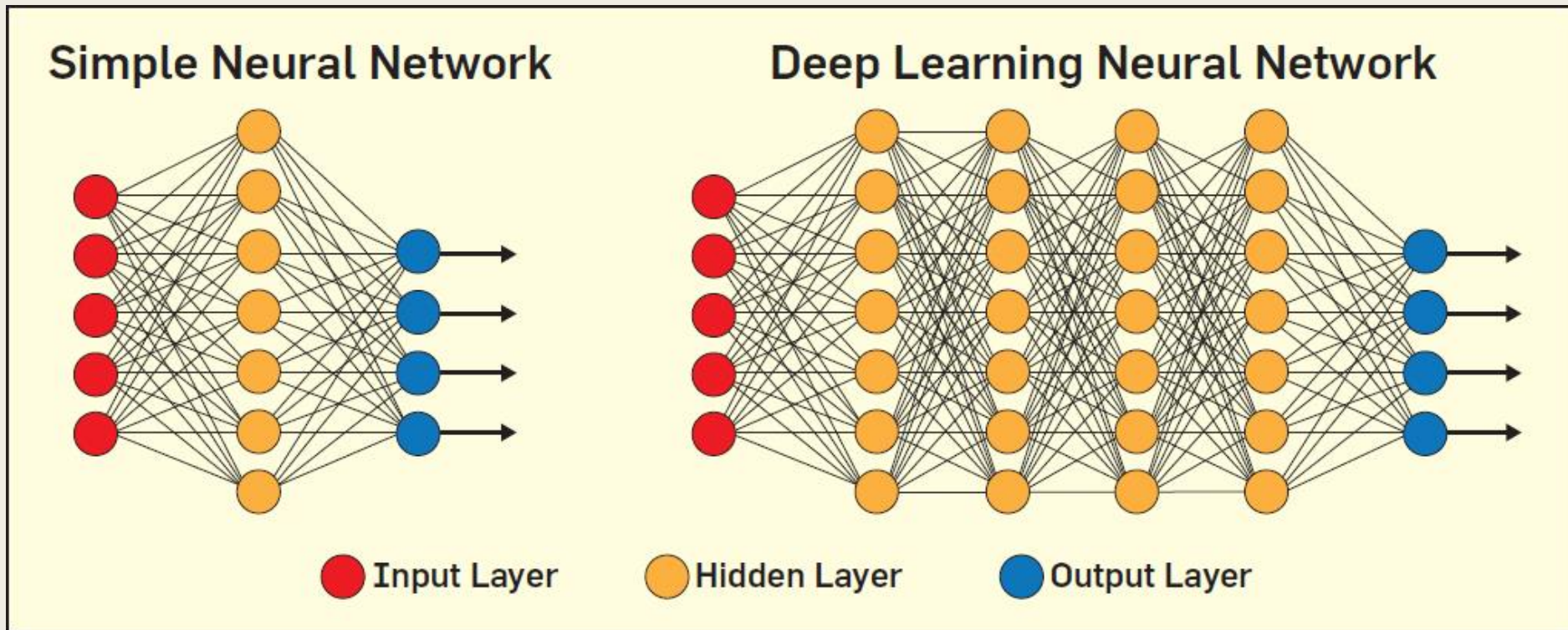
Recommendation System



Network Analysis



Deep Learning



ABOUT THIS COURSE

About the Course

- Overview
- Learning Objectives
- Readings
- Python Tutorials
- Assessment
- Grading
- Policies
- Schedule

About the Course

Learning Objectives

- Understand and Explain key machine learning paradigms beyond supervised learning, including clustering, dimensionality reduction, natural language processing, time series forecasting, network analysis, and deep learning.
- Apply and Implement advanced analytical techniques in Python to prepare data, build models, and evaluate performance across structured, unstructured, and temporal datasets.
- Analyze and Interpret model outputs to uncover hidden patterns, relationships, and insights that inform business and organizational decision-making.
- Integrate and Synthesize multiple machine learning methods—such as text, network, and deep learning approaches—to address complex, real-world problems.
- Communicate and Present analytical findings effectively through visualizations, reports, and presentations tailored to both technical and non-technical audiences.

About the Course

Readings

- McKinney, Wes (2023). Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter (3rd ed.). O'Reilly Media. [Free from Author](#) or [Amazon](#)
- VanderPlas, Jake (2022). Python Data Science Handbook: Essential Tools for Working with Data (2nd ed.). [Free from Author](#).
- Géron, Aurelien. (2025). Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems (3rd ed.). [O'Reilly Media](#).
- James, Gareth, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor (2023). An Introduction to Statistical Learning: With Applications in Python. Springer. [Free from Authors](#).
- Jurafsky, Daniel, and James H. Martin (2025). Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition. [Free from Authors](#).
- Bird, Steven, Ewan Klein, and Edward Loper (2021). Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit. [Free from Authors](#).
- Joseph, Manu and Tackes, Jeffrey (2023). Modern Time Series Forecasting with Python. Birmingham, UK: Packt Publishing. [Amazon](#).
- Chapman, Chris, and Elea Feit (2019). R for Marketing Research and Analytics. 2nd Edition. Springer International Publishing. [Amazon](#).
- Banik, Raunak (2018). Hands-On Recommendation Systems with Python: Start Building Powerful and Personalized Recommendation Engines with Python. Packt Publishing. [Amazon](#).
- Knickerbocker, D. (2023). Network science with Python: Explore the Networks Around Us Using Network Science, Social Network Analysis, and Machine Learning. Packt Publishing. [Amazon](#).

Python Tutorials

- Getting Started: [Kaggle](#)
- [LinkedIn Learning](#) Courses (free for Pace University Students)
- [DataCamp](#)

Assessment

- Assignments (200 points)
 - *Four assignments, each worth 25 points*
- In-class Quiz (100 points)
 - *Eight Quizzes, each worth 12.5 points*
- Project (300 points)
 - *Proposal: 50 points*
 - *Report: 200 points*
 - *Presentation: 50 points*
- Class Participation and Engagement (100 points)
- Exam (300 points)

This course is

for those who...

- have an open mind
- have a growth mindset
- are not intimidated by computer code
- are willing to spend many hours on an assignment
- are willing to work throughout the semester
- are willing to persist in the face of repeated failure

NOT for those who...

- have a fixed mindset
- are afraid of computer code
- have a packed work/course schedule
- tend to get frustrated easily
- prefer point and click menus like Excel
- fear math, statistics, or numbers

Student Expectations

- Attend class
- Be attentive in class
- Every student gets treated the same
- Due dates are not flexible
- No excuses
- Academic integrity





Schedule

- View detailed schedule on Syllabus

SOFTWARE

Analysis Tools

Programming Languages

- Python
- R
- Julia

BI Tools

- Tableau
- Power BI
- QlikView

Database

- MySQL
- PostgreSQL

Programming Languages

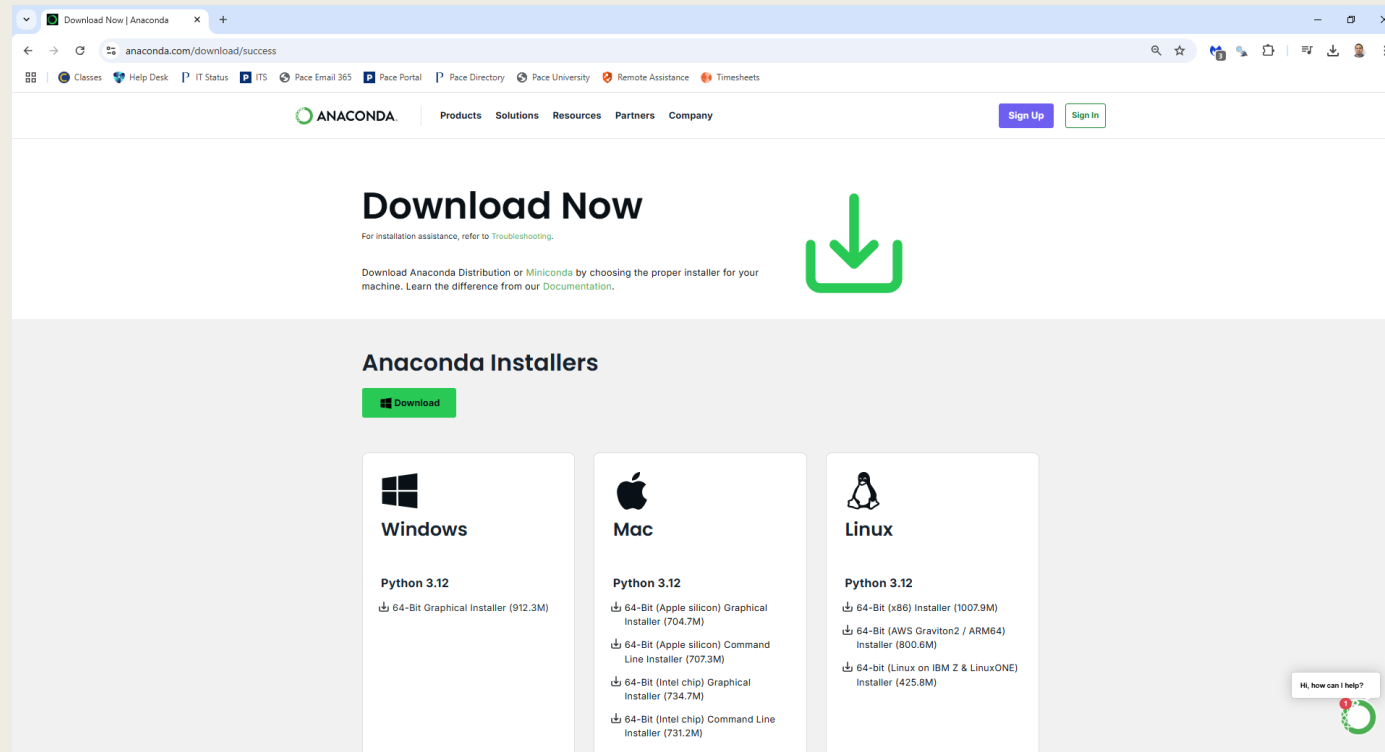
1. Python
2. R
3. SQL
4. Java
5. Julia
6. Scala
7. C/C++
8. JavaScript
9. Swift
10. Go
11. MATLAB
12. SAS

[Source: DataCamp](#)

Python

- Jupyter Notebook
 - [*Anaconda*](#)
 - [*Miniconda*](#)
 - [*Python*](#)
- [Google Colab](#)
- [VS Code](#)
- [Rstudio](#), Positron

Install Anaconda



- Installation Type: Just Me
- Choose Install Location: Default
- Advanced Installation Options: Use defaults

....while we wait



Student Introductions

- Name
- Educational background
- Analytics-related topics, software, or programming languages you are familiar with.

Setup Environment

- Create an environment for a project or set of related projects.
- Different projects may need different versions of Python or libraries
- Avoids version conflict between libraries
- Enables reproducibility. Code on the same environment will generate same results
- Manage environments with conda.
- Create and load environments by
 - *Command Line: Anaconda Prompt (in Windows) or Terminal (in Mac)*
 - *Anaconda Navigator*

Setup Environment (using Anaconda Prompt or Terminal)

■ Environment

- *Navigate to working folder. E.g.,*
- *> cd /myfiles/my_favorite_course*
- *> conda create --name ds_env python=3.13.5*
- *> conda activate ds_env*
- *> pip install ipykernel # Ensure ipykernel is installed in the environment*
- *> python -m ipykernel install --user --name ds_env --display-name "ds_env"*

■ Libraries

- *> conda install numpy pandas*
- *> jupyter notebook*

■ New Notebook

- *In browser window. New → ds_env*

Setup Environment using Anaconda

- Open Anaconda Navigator.
- Click on the "Environments" tab. Click "Create" at the bottom.
- Enter a name for your new environment (e.g., ds_env). Choose the Python version (select 3.13.5). Click "Create" to initialize the new environment. Once created, you can select the environment and install packages as needed.
- Setup environment once. Next time, initialize the environment from the Environments tab in Anaconda Navigator

Install JupyterLab and Libraries

- In Anaconda Navigator:

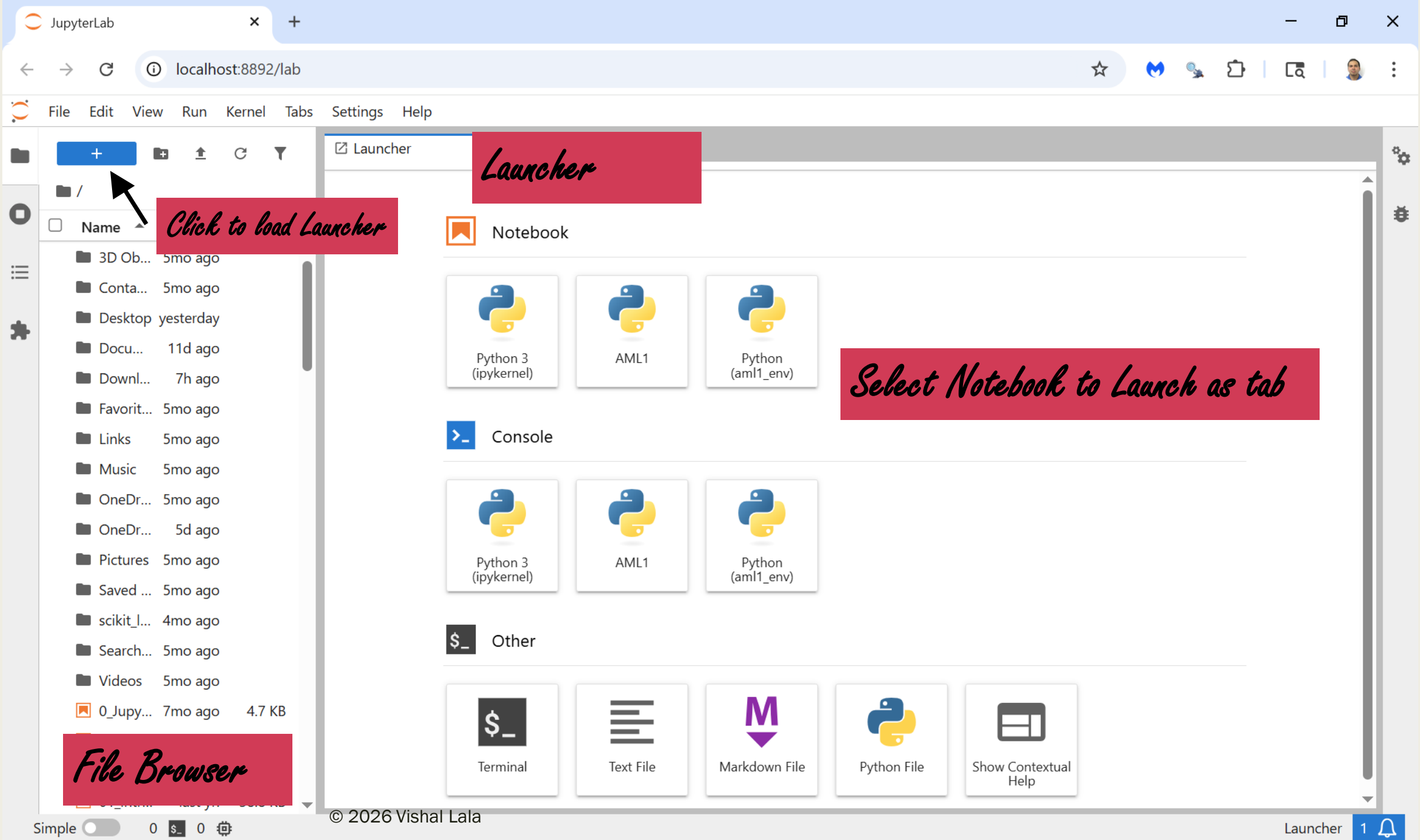
- Click on the "Environments" tab.
- Select your environment (e.g., `ds_env`) from the left sidebar.
- Use the dropdown to show "Not installed" packages.
- Search for JupyterLab, check the box next to it and click "Apply" to install it.
- Search for other libraries like numpy and pandas. Check the boxes next to it and click "Apply" to install them.

Launching JupyterLab

- Open Anaconda Navigator
 - *Launch JupyterLab from the list of applications in the Home tab.*
 - *Select Environment from drop down in pop-up window*

JupyterLab

- Launcher: Create notebook, console, terminal.
- Notebook Area: Write and run code. Multiple notebooks can be opened in tabs
- File Browser: Navigate project files.



Launcher

Click to load Launcher

Select Notebook to Launch as tab

File Browser

Jupyter Notebook

- Use Jupyter Notebook to manage Python code
- Key content types
 - *Markdown cell to add notes*
 - *Code cell to enter executable Python code*
 - *Results of running code appear below code cell*

Untitled32.i... - JupyterLab

localhost:8892/lab/tree/Untitled32.ipynb

File Edit View Run Kernel Tabs Settings Help

Untitled32.ipynb

Markdown

AML1

Use Markdown Cells for adding notes. Code cell below will run code and print output below the code cell

[1]: print('Use Code cells for running Python code')

Use Code cells for running Python code

[]:

Simple 3 3 AML1 | Idle © 2026 Vishal Lala

Mode: Edit Ln 1, Col 2 Untitled32.ipynb 1

Helpful Resources

- [Anaconda Navigator](#)
- [Jupyter Documentation](#)
- [Python Tutorials](#)

Summary

- In this module,
 - *Compared supervised learning to unsupervised and specialized machine learning techniques*
 - *Examined the structure and expectations of the course*
 - *Setup a Python environment*